

Automated Classification of Customer Messages for Credit Risk Monitoring

Naive Bayes Case Study - Business and Technical Report

Data Mining Project Report

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Note: Artificial intelligence was used in generating the code

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Executive Summary

This case study proposes a first version of an automated message triage system for retail banking. The system uses a Naive Bayes text classifier to distinguish routine customer messages from messages that may signal financial distress. The objective is to help the bank prioritize high-risk communications faster and reduce the chance that urgent credit-risk signals remain untreated.

1. Strategic Context

The bank receives large volumes of written customer messages through digital channels such as email, online contact forms, and mobile banking applications. Many messages are routine, but some include early signals of repayment difficulty or financial pressure. Without automation, all messages are handled with similar urgency, which can delay escalation of critical cases.

2. Mission Objective

The objective is to build a scalable, interpretable text classification model that assigns incoming messages to operational risk categories. In this implementation, the model separates messages into low_risk and high_risk. High-risk messages should be routed to credit risk officers for faster review, while low-risk messages can remain in the normal customer service queue.

3. Dataset and Label Mapping

The UCI SMS Spam Collection dataset is used as a public proxy dataset. The original labels are reinterpreted for the banking context: ham messages become low_risk messages, and spam messages become high_risk messages. This creates a practical binary classification task similar to screening customer communications for potential risk.

Original Label	Banking Interpretation	Operational Meaning
ham	low_risk	Routine information request or low urgency message
spam	high_risk	Proxy for financially risky or urgent communication

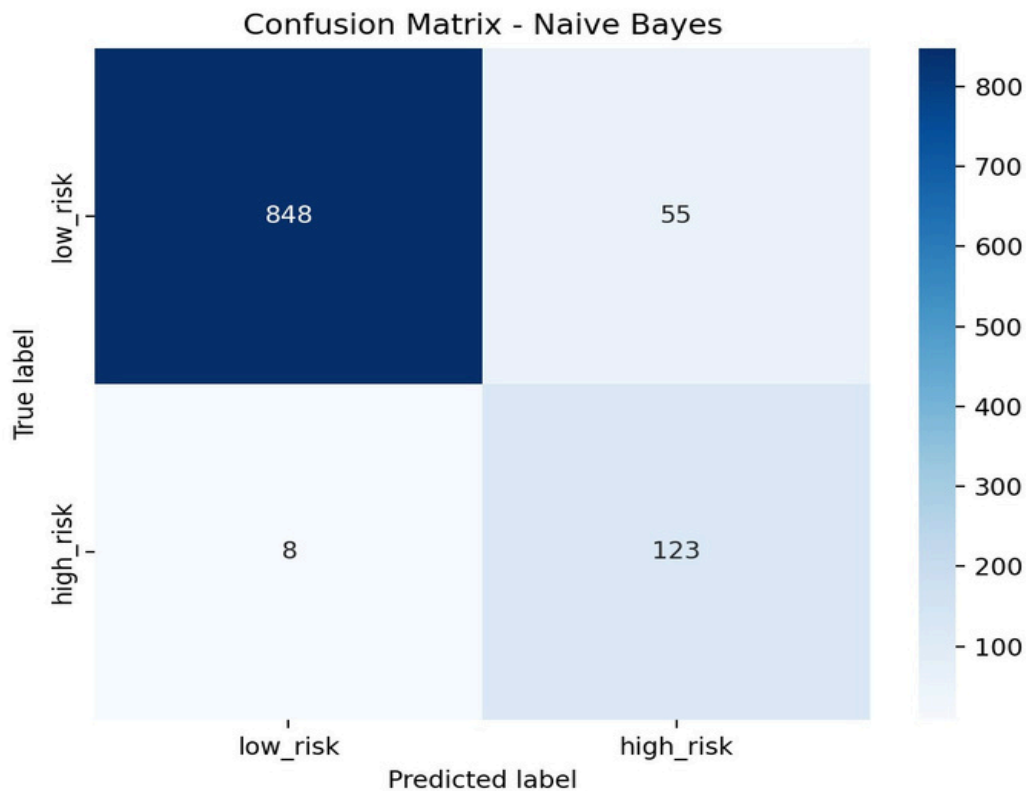
4. Methodology

The solution follows a simple, interpretable machine learning pipeline that can be explained to both technical and business stakeholders.

Step	Action	Purpose
1	Text cleaning	Lowercase text and remove URLs, numbers, punctuation, and extra spaces.
2	TF-IDF vectorization	Convert messages into numerical features based on important words and word pairs.
3	Train Naive Bayes	Use Complement Naive Bayes, a text-friendly Naive Bayes variant suited for imbalanced data.
4	Evaluate	Use accuracy, precision, recall, F1-score, and the confusion matrix.
5	Interpret	Translate technical results into business risk decisions.

5. Model Results from the Confusion Matrix

The confusion matrix below shows how the model performed on the test set. Rows represent the true labels and columns represent the predicted labels.



6. Business Interpretation of the Results

Metric	Value	Business Meaning
True low-risk predicted low-risk	848	Routine messages correctly kept in the normal queue.
Low-risk predicted high-risk	55	False positives: extra review workload.
High-risk predicted low-risk	8	False negatives: risky messages missed. This is the most dangerous error. Risky messages correctly escalated.
True high-risk predicted high-risk	123	Overall share of correct predictions.
Accuracy	93.91%	The model caught most high-risk messages.
High-risk recall	93.89%	Some escalated messages were actually low-risk.
High-risk precision	69.10%	

For this banking use case, the main priority is not only accuracy. The bank should pay close attention to recall for the high_risk class. A false positive creates extra review work, but a false negative can delay intervention with a customer who may be experiencing repayment difficulty.

Key interpretation: the model correctly identified 123 high-risk messages and missed only 8. This is a strong result for an early-warning triage tool. However, 55 low-risk messages were incorrectly escalated, so the model should be used as a prioritization assistant rather than a fully autonomous decision system.

7. Recommended Operational Workflow

Stage	Recommendation
1. Message intake	Every incoming digital message is automatically cleaned and vectorized.
2. Risk scoring	The Naive Bayes model predicts low_risk or high_risk and outputs a probability score.
3. Priority routing	Predicted high-risk messages are sent to a priority queue for credit risk officers.
4. Manual validation	Risk officers review flagged messages before any business decision is made.
5. Continuous improvement	Newly reviewed internal messages are added to the training set to improve future performance.

8. Limitations and Next Steps

The SMS dataset is only a proxy for banking messages, so the model should be validated on real internal customer communications before deployment. The current implementation is binary, while the mission brief also mentions contractual disputes. A future version should use a real banking dataset with three or more labels: financial distress, contractual dispute, and general inquiry.

Recommended next steps: collect labeled bank messages, add a multi-class model, tune the decision threshold to reduce false negatives, and integrate the scoring output into the bank CRM or customer service workflow.

Conclusion

This Naive Bayes system is a practical and interpretable first step toward automated credit-risk monitoring. It transforms unstructured customer text into structured risk indicators and helps the bank respond faster to potentially critical cases.